

Mach Zehnder Interferometer Design Proposal

Elizabeth Strehlow

estrehlo

GitHub fork: [estrehlo/openEBL-2026-10](https://github.com/estrehlo/openEBL-2026-10)

1. Introduction

The Mach Zehnder Interferometer (MZI) is an optical device used to determine the relative phase shift between two beams of light split from a single source. In silicon photonics, an MZI with deliberate path length differences (a difference in length between the two arms) can be used to extract the group index (n_g) of a waveguide. As the wavelength is swept, the phase difference of the arms varies resulting in an interference pattern in the output intensity. The period of the pattern is the free spectral range (FSR), which is dependent on the known path length difference (ΔL) and n_g . Therefore we can use the MZI to calculate n_g by measuring the output spectrum.

2. Theory

2.1. MZI Operation

The MZI contains an input graded coupler, a Y-branch splitter, two wave guide arms (with lengths L_1 and L_2), a Y-branch combiner, and an output graded coupler.

The light entering the MZI is divided equally by the splitter:

$$E_{i1} = E_{i2} = \frac{E_{in}}{\sqrt{2}}$$

The light then propagates independently down each arm, accumulating a phase determined by the length of the arm and the waveguide effective index (n_{eff}).

$$E_{o1} = E_{i1}e^{-i\beta L_1}$$

$$E_{o2} = E_{i2}e^{-i\beta L_2}$$

$$\beta(\lambda) = \frac{2\pi n_{eff}(\lambda)}{\lambda}$$

The two fields recombine at the combiner with an amplitude factor of $1/\sqrt{2}$ as shown in the following formula:

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2}(e^{-i\beta L_1} + e^{-i\beta L_2})$$

Where $I_i = |E_{in}|^2$ is the input intensity, the output intensity (I_o) is calculated as follows:

$$I_o = |E_o|^2 = \frac{I_i}{4}|e^{-i\beta L_1} + e^{-i\beta L_2}|^2 = \frac{I_i}{2}[1 + \cos(\beta\Delta L)]$$

An interesting case to consider is $\Delta L = 0$, where the arms of the MZI are of equal length. In that case, there is no loss in intensity detected at the output because the equal length arms result in equal phase light waves, so there is no destructive interference when they recombine.

$$I_o = \frac{I_i}{2}[1 + \cos(0)] = \frac{I_i}{2}[1 + 1] = \frac{I_i}{2}[2] = I_i$$

However, for an unbalanced interferometer, the period of the wave at the output, FSR, is calculated as follows, where the result is in meters:

$$FSR = \frac{\lambda^2}{n_g \Delta L}$$

2.2. Derivation of Group Index from FSR

Now we can obtain the group index, n_g , a quantity which is not directly measurable, using the resulting FSR from 2.1 along with the known values ΔL and λ :

$$n_g = \frac{\lambda^2}{FSR \cdot \Delta L}$$

3. Modeling and Simulation

3.1. Baseline Waveguide Characterization

For the experiments, there is a baseline device which is characterized, then experiments are carried out on two aspects of the device. For the baseline, the quasi-TE polarized waveguide has a width of 500 nm and a height of 220 nm. The base wavelength value is 1550 nm.

Modal analysis, performed in Lumerical MODE, confirmed single-mode TE behavior and was used to extract the fundamental mode's properties at 1550 nm.

Figure 1 shows the simulated electric field intensity for the fundamental mode of the baseline structure on a logarithmic scale. The profile shows the field is well-confined

and decays smoothly towards the edges. The mode was 98% TE-polarized with $n_{eff} = 2.44682$ and $n_g = 4.20368$ at 1550 nm wavelength.

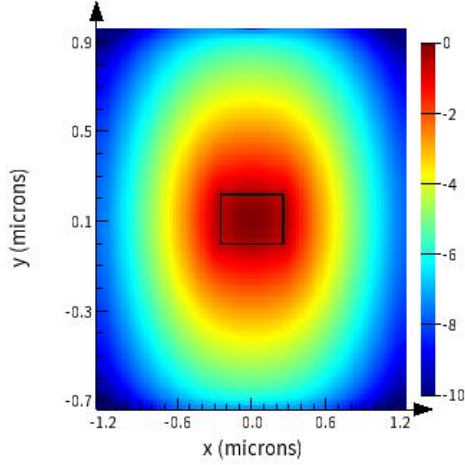


Figure 1. Electric field intensity for baseline device at 1550 nm wavelength.

A frequency sweep from 1500 to 1600 nm was performed to evaluate dispersion in the waveguide. Figure 2 shows the smooth decrease in n_{eff} across the frequencies while Figure 3 shows the group index smoothly increasing. These trends match the expected behavior of the waveguide.

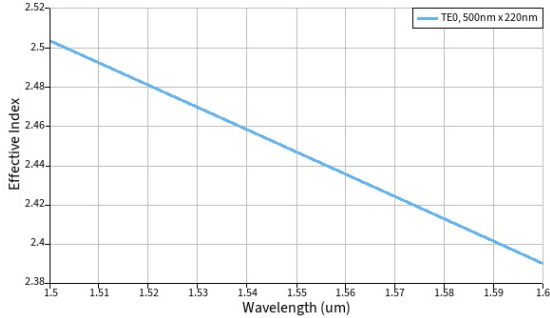


Figure 2. Effective group index vs wavelength for baseline device.

The simulated n_{eff} was fit to a second-order polynomial in Lumerical MODE, resulting in Figure 4. The fit line fitting tightly to the simulated data shows the model is performing as expected.

3.2. Experiment 1: Free Spectral Range vs Arm Length Imbalance(ΔL)

The purpose of the first experiment is to demonstrate the inverse relationship between the imbalance length of the interferometer and the free spectral range (FSR). To perform

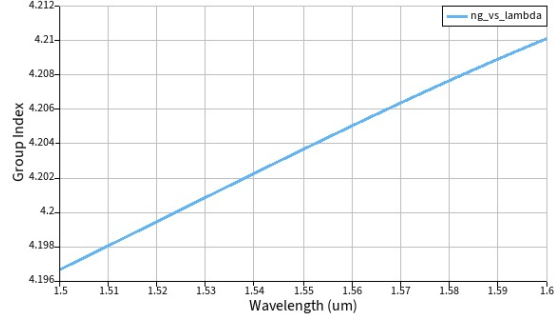


Figure 3. Group index vs wavelength for baseline device.

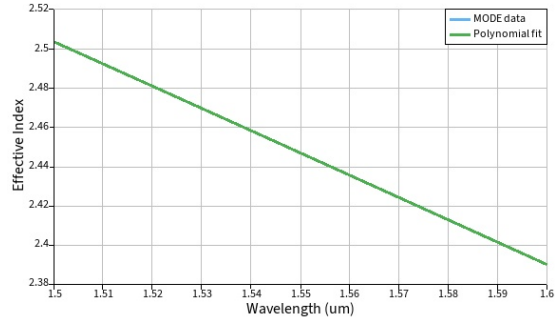


Figure 4. Polynomial fit of effective group index vs wavelength for baseline device.

this experiment, five MZIs were designed with varying differences in arm length. The output intensity values will be used to calculate the group index, which in turn can be used to find the FSR for each ΔL .

3.2.1 Fixed Parameters

For this experiment, the waveguide width is fixed at a value of 500 nm. The wavelength is swept from 1500 to 1600 nm.

3.2.2 Independent Variable

The independent variable for this experiment is the difference in arm lengths in the interferometer, δL . The relationship between ΔL and FSR is an inverse relationship, so the values selected for ΔL double from one value to the next. That is, they are geometrically spaced rather than being evenly spaced over the range. The selected values are 20 nm, 40 nm, 80 nm, 160 nm, and 320 nm.

3.2.3 Modeling and Simulation

Using the calculations in 2, the estimated FSR values are shown in Table 1.

Using Lumerical MODE, the light's output intensity is shown for each value of ΔL in Figure 5. This plot shows how the larger difference in arm lengths results in a higher

Table 1. Simulated FSR values varying ΔL .

ΔL (μm)	Simulated FSR (nm)
20	28.6
40	14.3
80	7.1
160	3.6
320	1.8

frequency of peaks and so the FSR will reduce as the arm length difference grows.

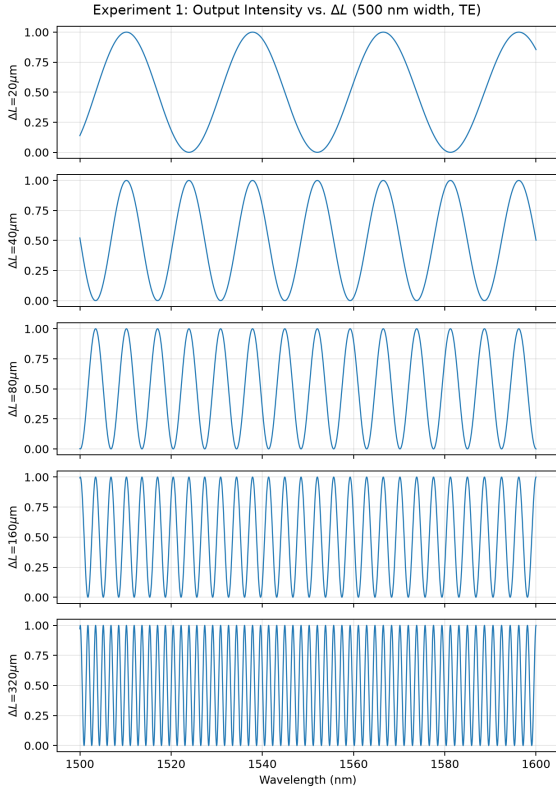


Figure 5. Output intensity vs ΔL for a 500 nm wide quasi-TE polarized MZI.

Figure 6 shows the curves from Figure 5 overlaid on top of each other, centered at 1550 nm wavelength.

Finally, using the effective group index can be calculated from the derived FSR values. This is calculated by looking at two adjacent intensity peaks, say for the curve for $\Delta L = 80$ nm. One set is measured at 1552.06 nm and 1559.25 nm, which is a simulated FSR value of 7.195 nm. The resulting $n_g = 4.2044$, which is 0.017% difference from 4.2037 calculated by the FDE dispersion calculation.

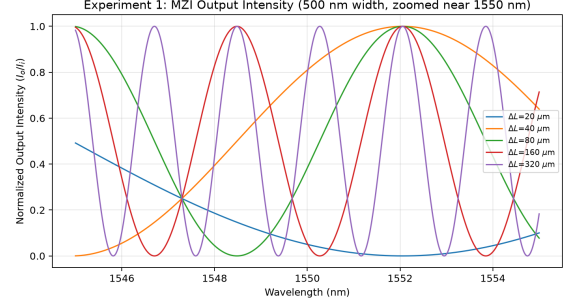


Figure 6. Overlay of output intensity vs ΔL for a 500 nm wide quasi-TE polarized MZI.

3.3. Experiment 2: Group Index vs Waveguide Width

Experiment 2 was created to demonstrate how the group index of the waveguide varies with the waveguide width. The expected result of this experiment is that n_g will decrease as the width of the waveguide increases.

3.3.1 Fixed Parameters

For this experiment, ΔL will remain fixed at 80 nm. The device will be quasi-TE polarized and the same topology for the splitter and combiners will be used with tapers to adjust the waveguide width to the 500 nm port width.

3.3.2 Independent Variable

The core waveguide width will be swept across 5 devices with widths of 400 nm, 450 nm, 500 nm, 550 nm, and 600 nm.

3.3.3 Modeling and Simulation

Figure 7 shows an example of the mode profiles for the 400 nm and 600 nm devices. This figure illustrates the confinement of the waveguide of the 400 nm (left) device versus the 600 nm (right) device.

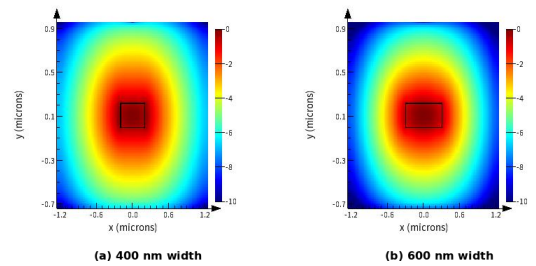


Figure 7. (l) Electric field intensity for 400 nm wide waveguide. (r) Electric field intensity for 600 nm wide waveguide.

Table 2 contains the simulated values for n_{eff} and FSR for each of the waveguide widths. The values for n_g

were obtained by running a frequency sweep in Lumerical MODE for each waveguide width. The frequencies swept ranged from 1500 to 1600 nm. As expected, n_g decreases as the width increases. This is also demonstrated in Figure 8. The FSR values were then calculated from the formula in 2.2. Since n_g and FSR have an inverse relationship, FSR is seen to increase along with the waveguide width, as shown in Figure ??.

Table 2. Simulated n_g values varying waveguide width.

Width (nm)	n_g	FSR (nm)
400	4.4008	6.82
450	4.2992	6.99
500	4.2037	7.15
550	4.1252	7.28
600	4.0625	7.39

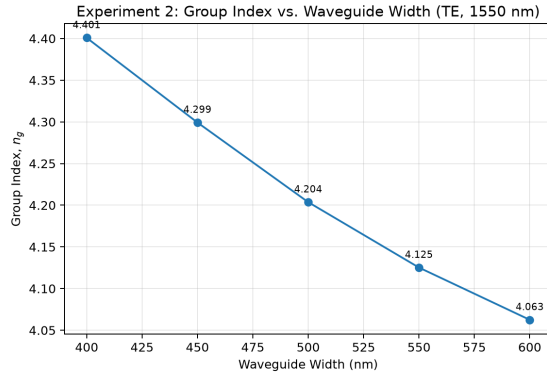


Figure 8. Group index vs waveguide width for $\Delta L = 80$ nm.

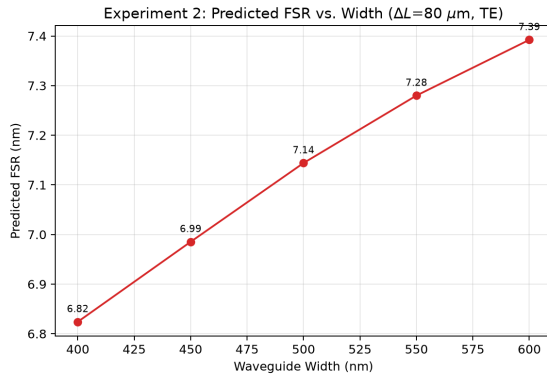


Figure 9. FSR vs waveguide width for $\Delta L = 80$ nm.

4. Layout

For the layout, the graded couplers are required to follow a $127 \mu m$ pitch for the automated measurement tool to align to the input and output graded couplers. The bend radius

used the default minimum of $5 \mu m$, which ensures the bends are manufacturable and will not degrade the light flowing through the curves.

In order to fit the large number of devices for test, the two largest devices in Experiment 1 required serpentine shaped waveguides to fit the longer length.

In total, there are ten devices, with the baseline device included in both experiments, so it is a copy of itself.

5. Summary and Next Steps

With both experiments designed and simulated, the next steps are to finalize the layout for fabrication, which include completing the layout for Experiment 2. There is no taper contained in our library which fits what the experiment requires, so I will need to create that structure.

The measured data collected by the instructors will be used to extract experimental FSR and group index values as described in 2. Then that will be compared to the simulated values presented here so that we can identify and discuss mechanisms leading to discrepancies (i.e. loss).